# AWS VPC Peering Project

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## 1. Objective

Connect EC2 workloads in VPC A and VPC B privately through VPC peering, using explicit routes and appropriately scoped security rules. The EWB workbook additionally specifies a public bastion host for SSH administration.

## 2. Observed AWS resources

VPC A: 10.10.0.0/16; VPC B: 10.20.0.0/16. Screenshots show two running t3.micro EC2 instances named VPC-A-Test-Server and VPC-B-Test-Server. A Session Manager terminal shows private IP 10.10.147.115 on the VPC A instance. The successful ping target is 10.20.2.43 in VPC B. These actual IPs differ from the workbook example.

## 3. Routing

The observed VPC B main route table includes 10.10.0.0/16 to a VPC peering connection and a local 10.20.0.0/16 route. VPC A private route table screenshots show 10.20.0.0/16 directed toward a peering connection. Both directions require valid routes, subnet associations and security-group permissions.

## 4. Connectivity test observed

The VPC A Session Manager terminal shows ping -c 4 10.20.2.43 with four received packets and 0% packet loss. This demonstrates successful ICMP reachability from the displayed VPC A instance to the displayed private destination at the time of the screenshot.

## 5. Supporting setup

Screenshots also show IAM instanceRole with AmazonSSMManagedInstanceCore and CloudWatchAgentServerPolicy, SSM/SSMMessages interface endpoints, and security-group settings. These are implementation details; VPC endpoints are not a substitute for the workbook-required bastion SSH demonstration.

## 6. Evidence gaps / pending verification

The supplied screenshots do not establish a bastion host, SSH ProxyJump, a pre-peering baseline failure, reverse ping, controlled ICMP failure and recovery, or cleanup. These tests must be performed and documented if the full workbook requirements apply. Do not represent them as completed.

## 7. Learning outcomes

VPC peering provides private connectivity but is not transitive. Routing is required in both directions. Security-group rules control permitted traffic. Private hosts can communicate without receiving public IPv4 addresses.

## 8. Responsible cleanup

After capturing evidence, remove project resources no longer needed and review AWS Billing. Preserve cleanup screenshots for the final report. Do not upload SSH private keys or credentials.

## 9. Reference

EWB Courses, AWS VPC Peering Student Project Workbook, October 2026, pages 2-14.